

$$\tilde{C} = \begin{pmatrix} C_{11} & C_{12} & \cdots & C_{1m} \\ C_{12} & C_{22} & \cdots & C_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1m} & C_{2m} & \cdots & C_{mm} \end{pmatrix} \quad C_{ij} = \begin{pmatrix} \sigma_{ij} & 0 & \cdots & 0 \\ 0 & \sigma_{ij} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_{ij} \end{pmatrix}$$

Please note that $\tilde{C}$ is a symmetric matrix, therefore we have, $C_{ij} = C_{ji}$ for all i and j .

The equality constraint matrix $\tilde{A}_1$ and vector $\tilde{b}_1$ are:

$$\tilde{A}_1 = \begin{pmatrix} \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 1 \end{bmatrix} & \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \end{bmatrix} & \cdots & \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \end{bmatrix} \\ \begin{bmatrix} 0 & 0 & \cdots & 1 \\ 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix} & \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \end{bmatrix} & \cdots & \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ 0 & 0 & \cdots & 1 \end{bmatrix} \end{pmatrix} \quad \tilde{b}_1 = \begin{pmatrix} S_1 \\ S_2 \\ \vdots \\ S_m \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

The inequality constraint matrix $\tilde{A}_2$ and vector $\tilde{b}_2$ are:

$$\tilde{A}_2 = \begin{pmatrix} \begin{bmatrix} A'_2 & 0 & \cdots & 0 \\ 0 & A'_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A'_2 \\ A''_2 & 0 & \cdots & 0 \\ 0 & A''_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A''_2 \end{bmatrix} \end{pmatrix} \quad A'_2 = \begin{pmatrix} -1 & 1 & 0 & \cdots & 0 \\ 0 & -1 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & -1 & 1 \end{pmatrix} \\ A''_2 = \begin{pmatrix} 1 & -1 & 0 & \cdots & 0 \\ 0 & 1 & -1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & -1 \end{pmatrix}$$

$$\tilde{b}_2 = \begin{pmatrix} \begin{bmatrix} b'_1 \\ b'_2 \\ \vdots \\ b'_m \\ b''_1 \\ b''_2 \\ \vdots \\ b''_m \end{bmatrix} \end{pmatrix} \quad b'_i = \begin{bmatrix} -x_i^* \\ -x_i^* \\ \vdots \\ -x_i^* \end{bmatrix} \\ b''_i = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

And x_i^* is the maximum quantity that can be traded in any period for stock i defined by the trader.